Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Using the limit definition of the derivative, find the derivative of the function $f(x) = \frac{x-1}{x}$. Find the equation of the tangent line to $f(x)$ at $x = 3$.

Exercise 2.

Find the derivative of the following expressions.

1. $5x^4 - \frac{2}{\sqrt[3]{x}} + x^e - e^x$.
2. $x^{3/2}\cos(x)$.
3. $e^x x^{3/2}\cos(x)$.
4. $\frac{\cos(x)}{e^x}$.

Exercise 3.

Find the equation of the tangent line to $g(x) = \frac{x^2 e^x}{e^x+1}$ at $x = \ln(2)$.

Exercise 4.

Consider the function $h(x) = x^{1/3}$.

1. Find the domain and the range of $h(x)$.
2. Find $h'(x)$, its domain, and its range.
3. Find the vertical asymptotes of $h'(x)$, and compute the one-sided limits on each side of the asymptotes.
4. Sketch the graph of $h(x)$. Draw the tangent line to $h(x)$ at $x = 0$. What is the slope of that line? Explain the relation between that slope and the asymptotes above.

Exercise 5.

Not every function has a derivative at every point. Consider the function $j(x) = |x|$, we will now try to find $j'(0)$.

1. Using the limit definition of the derivative, explain why $j'(0) = \lim_{x\to 0} \frac{j(x)-j(0)}{x}$.
2. Compute $j'(0) = \lim_{x\to 0} \frac{j(x)-j(0)}{x}$. If the limit does not exist, justify your answer.
3. Graph $j(x)$.
4. What happens when you try to draw a tangent line to $j(x)$ at $x = 0$?
5. Based on the above, what can you conclude about $j'(0)$?